GMR Institute of Technology

An Autonomous Institute Affiliated to JNTUGV, Vizianagaram

GARIT
Training Tomorrow's Engineers Today

V sem
PSP SEC-B
Dr. MRB

STUDENTS ATTENDANCE REGISTER

THEORY

Department of EEE

Name of the Faculty : Dr. M. RAMBABU Dept. EEE

Course Name : POWER SYSTEM PROTECTION Code 21EE504

Semester 5th Section B Academic Year 2024-25

GMR Nagar, Rajam - 532 127
Vizianagaram Dist., Andhra Pradesh, India

21EE504 Power System Protection

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Course Outcomes

1. Outline the working of various circuit breakers
2. Summarize the construction and working of different types of relays
3. Identify suitable protective device for power system equipment
4. Identify protection schemes for relaying equipment
5. Outline the operation of electrostatic and digital relays
6. Apply digital logic in relay communications

COs – POs Mapping

COs	PO_2	PO_3	PSO_2
1	2	2	1
2	2	2	1
3	3	3	2
4	3	3	2
5	2	2	1
6	3	3	2

3–Strongly linked | 2–Moderately linked | 1–Weakly linked

Unit I

Circuit Breakers

Basics of Protection & its significance, Circuit Breakers: Elementary principles of arc interruption, Restriking and Recovery voltages - Restriking Phenomenon, Average and Max. RRRV- Current Chopping and Resistance Switching - CB ratings and Specifications, Auto-reclosures, Description and Operation of Oil Circuit breakers, Air Blast Circuit Breakers, Vacuum Circuit Breakers and SF6 circuit breakers, MCB and MCCBs, Isolators.
Difference between a Fuse, an Isolator and a Circuit breaker.

12 Hours

Unit II

Electromagnetic Relays

Principle of operation and construction of attracted armature, Balanced beam, induction disc and induction cup relays. Instantaneous, DMT and IDMT relays.
Over current/ Under-voltage relays, Directional relays, Differential and percentage differential relays, Translay relay, Universal torque equation,
Distance relays- Impedance, Reactance and Mho relays.
Fundamental requirements of protective relays, Types of protection.

12 Hours

Unit III

Power system components protection

Generator Protection-Protection of generators against stator faults, rotor faults, restricted earth fault and inter-turn fault.
Transformer protection - Percentage differential protection, Buchholz relay protection.
Line protection -Over current, carrier current and three-zone distance relay protection using impedance relays, ZnO and rod gap Lightning arresters, grounding wires, Peterson coil.
Bus bar protection – Differential protection.
Voltage surge, lightning.

12 Hours

Unit IV

Static & Digital Relays

Static Relays-Introduction, Static relay components, Comparators – Amplitude and phase, Static over current relay, Static distance relay and Static poly-phase relay.
Digital Relays- Introduction, Digital logic communication, Microprocessor based over current, impedance, reactance & Mho relays, relay testing, static relays versus electromagnetic relays
Static relays versus electromagnetic relays.

12 Hours
Total: 48 Hours

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Name of the Course POWER SYSTEM PROTECTION

Name of the Faculty Dr. M. RAMBABU

LESSON PLAN AND DIARY

LESSON PLAN		DIARY			
Contact Hr. No.	Topic (s) Planned	Sl. No.	Date	Topic (s) Covered	Remarks
	UNIT-I			UNIT-I	
1.	Introduction to power system protection.	1	30/7/24	Introduction to power system protection.	
2.	Elementary principles of arc interruption,	2	1/8/24	Arc interruption process	
3.	Restriking voltage and Restriking phenomenon and recovery voltages.	3	2/8/24	Restriking voltage & Technical Terms.	
		4.	5/8/24	Slepain's Theory	
4.	Slepain's Theory	5.	6/8/24	Cassie's theory.	
5.	Cassie's Theory, RRRV	6	8/8/24	Resistance switching &	
6.	current chopping & Resistance switching.			current chopping.	
		7.	12/8/24	Restriking voltage & recovery voltage	
7	CB Ratings & specifications				
8.	Auto Reclosures.	8	13/8/24	CB ratings & specifications	
8.	Bulk oil circuit Breakers	9	19/8/24	Expression for RRRV	
9.	minimum oil CB.	10.	20/8/24	problems on RRRV.	
10	Air Blast circuit Breaker.	11	22/8/24	Bulk oil circuit Breaker	
11	Vaccum oil circuit Breaker.	12	26/8/24	Air Blast circuit Breaker	
12	SF6, circuit Breaker.	13	27/8/24	SF6, circuit breaker	
12	MCB, MCCBs and Isolators its working.	14	29/8/24	vacuum circuit Breakers	
		15	2/9/24	MCB, MCCB and Isolators its working.	
	UNIT-II.			UNIT-II	
13.	Attracted Armature, solenoid type, Balanced Beam	16.	3/9/24	Attracted Armature.	
14.	Induction Type Relays.	17	10/9/24	Induction type Relays	
15.	Instantaneous, DMT, IDMT, Type Relays.	18	12/9/24	DMT, IDMT, Relays.	
		19	17/9/24	over current Relays	
16.	over current Relays.	20	19/9/24	under voltage Relay	
17.	under voltage Relays.	21	20/9/24	Directional power relay.	
18.	Directional power Relays.	22	23/9/24	Differential Relays	
19.	Differential Relays.	23	24/9/24	Translay Relay &	
20.	Translay relay.			universal Torque equation.	
21	universal Torque equation.	24	26/9/24	Distance Relays -	
22	Distance Relays - impedance its working			impedance its working	
		25	28/9/24	Reactance Relays operation	
23	Reactance Relays operation	26	7/10/24	mho Relays and its operation.	
24.	Mho Relays and its operation.				

LESSON PLAN AND DIARY

LESSON PLAN		DIARY			
Contact Hr. No.	Topic (s) Planned	Sl. No.	Date	Topic (s) Covered	Remarks
	UNIT-III			UNIT-III	
25.	Generator protection.	27	8/10/24	Generator faults & protection.	
26.	Stator faults of Generator.				
27.	Rotor faults.	28	10/10/24	Stator faults & rotor faults.	
28.	Restricted Earth fault and Inter turn fault protection.	29	15/10/24	Restricted Earth fault protection.	
29.	Types of faults in Transformer, Differential protection	30	16/10/24	Inter turn fault protection.	
		31	17/10/24	Types of faults in Transformer & its protection.	
30.	Buchholz Relay protection.				
31.	Line protection from over current fault.	32	21/10/24	Buchholz protection	
		33	22/10/24	Line protection from over current fault	
32	Carrier current protection.				
33	Three zone distance Relay protection using impedance Relays.	34	24/10/24	Three zone distance protection	
		35	28/10/24	Bus bar protection	
34.	ZnO and Rod gap Lightning arrester.	36	29/10/24	ZnO & Rod gap Lightning arresters.	
35.	Grounding wire and peterson coil impedance.	37	5/11/24	Groundwire & peterson coil Grounding.	
36.	Busbar protection Differential protection	~~38~~	~~7/11/24~~	UNIT-IV	
		38	7/11/24	Static & Digital Relays.	
37.	UNIT-IV.	39	11/11/24	Relay components & its operation.	
37.	Static & Digital Relays.	40	11/11/24	Comparators - Amplitude & phase comparator.	
38.	Static Relay components its operation.	41	12/11/24	Static over current Relay	
39	Comparators - Amplitude and phase impedance	42	12/11/24	Static distance Relay	
		43.	13/11/24	Digital logic communication.	
40	Static over current Relay working.	44	14/11/24	Direct Relay to Relay Digital logic communication.	
41.	Static distance Relay and polyphase Relay	45	16/11/24	Microprocessor Based over current Relay.	
42	Digital logic communication and its role in protection	46	18/11/24	Microprocessor based impedance Relay & Reactance Relay	
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43.	Direct Relay to Relay Digital logic communication	47	19/11/24	Microprocessor based MHO Relay.	

LESSON PLAN AND DIARY

LESSON PLAN		DIARY			
Contact Hr. No.	Topic (s) Planned	Sl. No.	Date	Topic (s) Covered	Remarks
44.	microprocessor based over current relay	48	20/11/24	comparison of static & electromechanical relays & relay testing	
45.	Impedance relay				
46.	Reactance relay				
47.	mho relay				
48.	comparison of static & Electromechanical relay				
	Relay testing				